DENSITY OF BOUNDARY ENTROPY SPECTRA FOR SURFACE GROUPS

ABSTRACT. Let Γ_g be a closed orientable surface group of genus $g \geq 2$, and let ϑ be a finitely supported probability measure whose support generates Γ_g as a semigroup. We prove that its boundary entropy spectrum is dense in $[0, h_{\mathrm{RW}}(\Gamma_g, \vartheta)]$. The approximants are essentially free projective boundaries of faithful representations over the fixed field $\mathbb{Q}(i, \sqrt{2})$; their matrix images may be chosen dense in $\mathrm{SL}_2(\mathbb{C})$, with stationary measures of full support and exact dimension two. For the uniform eight-step walk on the genus-two group, a comeager set of targets is attained exactly, and the entire interval $[0, \frac{1}{2}\log 3]$ is attained. The proofs combine an explicit rational representation curve, Galois approximation, stationary-measure dimension formulas, and a category argument using bifurcation currents. Applications include full realization for coordinate products and continuum many nonisomorphic proper boundaries. We also give a self-contained relative entropy inequality for joins, a complete metric on the boundary lattice, and an explicit discontinuity of representation entropy. Full realization for the original genus-two walk remains open in this paper.

1. INTRODUCTION

A Poisson boundary records the asymptotic information in a random walk. Its measurable factors retain different amounts of that information, measured by Furstenberg entropy. The boundary entropy realization problem asks which amounts can occur. Even when the group and its increment law are fixed, the distinction between approximating an entropy value and realizing it exactly is substantial.

Let G be a countable group and let ϑ be a probability measure whose support generates G as a semigroup. A ϑ-boundary means a measurable G-equivariant factor of the Poisson boundary of (G, ϑ), equipped with its stationary probability measure. For a stationary G-space (B, ν) we use the convention

$$h_\vartheta(B, \nu) = \sum_{g \in G} \vartheta(g) D(g_*\nu \| \nu), \qquad D(\alpha \| \beta) = \int \log \frac{d\alpha}{d\beta}\, d\alpha.$$

This agrees, by change of variables, with $-\sum_g \vartheta(g) \int \log(d(g_*^{-1}\nu)/d\nu)\, d\nu$. Write

$$h_{\mathrm{RW}}(G, \vartheta) = \lim_{n \to \infty} \frac{1}{n} H(\vartheta^{*n}), \qquad \mathscr{H}_\partial(G, \vartheta) = \{h_\vartheta(B, \nu) : (B, \nu) \text{ is a } \vartheta\text{-boundary}\}.$$

All logarithms are natural. For finite Shannon entropy, every boundary entropy lies between zero and h_{RW}, and the trivial and Poisson boundaries attain the endpoints; see [14, 13].

Our principal result concerns the full closure of this spectrum, with additional control of the boundaries realizing the approximating values.

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Theorem A (Density for surface groups)**.** *Let Γ_g be the fundamental group of a closed orientable surface of genus $g \geq 2$, and let ϑ be a finitely supported probability measure whose support generates Γ_g as a semigroup. Then*

$$\overline{\mathscr{H}_\partial(\Gamma_g, \vartheta)} = [0, h_{\mathrm{RW}}(\Gamma_g, \vartheta)].$$

Every value in this interval can be approximated by the entropies of essentially free projective boundaries associated to projectively faithful representations

$$\Gamma_g \longrightarrow \mathrm{SL}_2(\mathbb{Q}(i, \sqrt{2})).$$

The representations may be chosen with dense image in $\mathrm{SL}_2(\mathbb{C})$, and their stationary measures on $\mathbb{CP}^1$ may be chosen with full support and exact dimension two.

The measure ϑ remains fixed throughout this approximation. The proof interpolates a Lyapunov exponent along a rational representation curve connecting the trivial representation to a cocompact Fuchsian representation. Galois conjugation supplies a dense set of faithful parameters over one fixed number field. At these parameters, the dimension theorem of Rapaport–Ren [15] identifies boundary entropy with the smaller of h_{RW} and four times the Lyapunov exponent. Finite-index restriction extends the construction from genus two to every genus.

For exact realization we specialize to

$$\Gamma = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = e \rangle, \qquad \mu = \frac{1}{8} \sum_{s \in S} \delta_s, \quad S = \{a_1^{\pm 1}, b_1^{\pm 1}, a_2^{\pm 1}, b_2^{\pm 1}\}. \tag{1.1}$$

Set $h = h_{\mathrm{RW}}(\Gamma, \mu)$.

Theorem B (Generic and interval realization)**.** *The spectrum $\mathscr{H}_\partial(\Gamma, \mu)$ contains a dense G_δ subset of $(0, h)$, all of whose values are attained by essentially free projective boundaries. Moreover,*

$$[0, \tfrac{1}{2} \log 3] \subseteq \mathscr{H}_\partial(\Gamma, \mu). \tag{1.2}$$

In particular $[0, c_* h] \subseteq \mathscr{H}_\partial(\Gamma, \mu)$, where $c_* = 2 \log 3 / (3 \log 7) = 0.376383356\ldots$.

The generic statement follows from lower semicontinuity of entropy and the support theorem for bifurcation currents of Deroin–Dujardin [5]. A category argument selects points on a comeager set of exact Lyapunov levels. The interval assertion then uses a quotient onto $F_2 \times F_2$ and the actual coordinate-update measure induced by (1.1). Those quotient boundaries have a nontrivial kernel as Γ-spaces. Theorem 10.1 shows more generally that two comeager boundary entropy spectra give full realization for a coordinate product.

The normalization in Theorem B follows from

$$\tfrac{1}{2} \log 3 \leq h \leq \tfrac{3}{4} \log 7. \tag{1.3}$$

Indeed F_4 maps onto Γ preserving the eight labeled increments, whereas $(a_1, b_1, a_2, b_2) \mapsto (x, y, y, x)$ maps Γ onto F_2 and sends μ to its simple random walk. Entropy decreases under quotients. For the simple random walk on F_r, reduced length has speed $1 - 1/r$, and the endpoint conditioned on length is uniform on its sphere; hence its entropy is $(1 - 1/r) \log(2r - 1)$.

A separate part of the paper concerns selection of limits in the lattice of factors of one Poisson boundary. Its joins and meets are the generated and intersected completed invariant σ-algebras, respectively. We write $\mathsf{h}(A) = h_\mu(A)$ in the following statement.

Theorem C (Relative joins and entropy distance)**.** *For the fixed walk* (1.1)*, if D is a common factor of boundaries A and C, then*

$$\mathsf{h}(A \vee C) + \mathsf{h}(D) \le \mathsf{h}(A) + \mathsf{h}(C).$$

The formula

$$d(A, C) = 2\mathsf{h}(A \vee C) - \mathsf{h}(A) - \mathsf{h}(C)$$

defines a complete metric on the boundary lattice modulo null sets. Metric convergence implies strong convergence of conditional expectations. If $\mathsf{h}(A_n) \to t$ and $\sum_n [\mathsf{h}(A_n \vee A_{n+1}) - \mathsf{h}(A_n)] < \infty$, then

$$\mathsf{h}\left(\bigwedge_N \bigvee_{n \ge N} A_n \right) = t.$$

The proof of the relative join inequality uses the nonstationary entropy functional and convolution identity of Gorodetski–Kleptsyn [10]. We include the argument because the common-factor subtraction is useful for the selection criterion. Completeness alone does not establish closedness of the scalar entropy spectrum.

1.1. **Relation to previous work.** Dodds–Furman [7, Theorem A] prove a nontrivial interval of boundary entropies for surface groups. Their product theorem assumes genus at least three [7, Theorem B]; their continuity theorem imposes L^2 conditions on a stationary density and its logarithm [7, Theorem 4.1]. The latter hypotheses do not cover the circle-supported Fuchsian endpoint used here. Hochman–Solomyak [12] and Rapaport–Ren [15] instead relate entropy to stationary-measure dimension under arithmetic separation. We use these formulas at dense algebraic parameters and interpolate only the Lyapunov exponent, whose continuity follows from Bocker-Neto and Viana [2].

The lattice of Poisson-boundary factors and entropy continuity along monotone lattice limits are studied by Dodds [6]. Tamuz–Zheng [17, Section 4] use ordinary join subadditivity in their work on asymptotic entropy spectra. The relative inequality above and its metric consequences are presented with self-contained proofs. Brieussel–Oppelmayer [3] prove continuity for Bowen–Poisson bundles; membership in that class of stationary actions does not itself identify an action as a factor of the fixed Poisson boundary considered here. We also distinguish monotone factor convergence from a general compactness assertion; the relevant withdrawn claims in [1, 4] are discussed in Section 7.

1.2. **Scope and organization.** Theorem A excludes open entropy gaps for all walks in its statement. Theorem B leaves at most a meager set of missing values for (1.1); it does not establish full realization or the interval $[0, h/2]$. Theorem 11.1 gives a discontinuity of entropy on the faithful non-elementary representation locus. The level-set results in Sections 4 and 12 explain why parameter-space saturation alone does not settle every target.

Section 2 constructs the rational family, and Section 3 proves Theorem A. Sections 4–6 establish generic realization. The lattice results precede the applications in Section 10, which include (1.2) and full realization for coordinate products. The final sections give the discontinuity example, the remaining levelwise obstructions, and precise open selection questions.

2. A rational family and arithmetic faithfulness

For $z \in \mathbb{C} \setminus \{-1/2\}$ set

$$a(z) = 1 + 2z, \qquad d(z) = \frac{1 + 2z^2}{1 + 2z}, \qquad A(z) = \begin{pmatrix} a(z) & 2z \\ z & d(z) \end{pmatrix}, \qquad B(z) = \begin{pmatrix} a(z) & -2z \\ -z & d(z) \end{pmatrix}.$$

Both matrices have determinant one. Define

$$K(z) = [A(z), B(z)], \qquad C(z) = (1 + 14z)I + zK(z),$$

and the parameter domain

$$D = \{z \in \mathbb{C} : z \neq -1/2,\ \det C(z) \neq 0\}.$$

Since $\det C(z)$ is a rational function that is nonzero at $z = 0$, its zero set is finite. Thus D is the complex plane minus finitely many points and in particular is path connected. For every $z \in D$ define a representation by

(2.1)
$$\rho_z(a_1) = A(z), \quad \rho_z(b_1) = B(z), \quad \rho_z(a_2) = C(z)B(z)C(z)^{-1}, \quad \rho_z(b_2) = C(z)A(z)C(z)^{-1}.$$

This is a rational family defined over $\mathbb{Q}$. In fact $C(z)$ commutes with $K(z)$, and hence the relator evaluates to

$$K(z)\, C(z)[B(z), A(z)]C(z)^{-1} = K(z)C(z)K(z)^{-1}C(z)^{-1} = I.$$

Lemma 2.1 (Trivial and Fuchsian endpoints)**.** *Both $0, 1$ belong to D. The representation ρ_0 is trivial, and the projectivization of ρ_1 is a faithful cocompact Fuchsian representation of the genus-two surface group.*

Proof. At zero all four images equal I, and $C(0) = I$. At one the exact matrix calculations are

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}, \quad K = \begin{pmatrix} -15 & -56 \\ -4 & -15 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -56 \\ -4 & 0 \end{pmatrix}.$$

In particular $\det C = -224 \neq 0$ and

$$(\operatorname{tr} A, \operatorname{tr} B, \operatorname{tr} AB) = (4, 4, 6), \qquad \operatorname{tr}[A, B] = -30 < -2.$$

The Fricke characterization of one-holed-torus holonomy [9, Theorem 4.1.2 and Section 4.4] says that this pair is the holonomy of a hyperbolic one-holed torus with geodesic boundary.

Let ℓ be the axis of the hyperbolic element K. The matrix $C = K - \frac{1}{2}\operatorname{tr}(K)I$ has the same eigenlines as K, opposite nonzero eigenvalues, and negative real determinant. Its projective action, interpreted as an orientation reversing isometry of $\mathbb{H}^2$, is reflection in ℓ. To check the last assertion, conjugate the two eigenlines to $0, \infty$: the projective matrix becomes $\operatorname{diag}(1, -1)$, whose orientation reversing extension is $w \mapsto -\overline{w}$, reflection in the imaginary axis.

Double the one-holed torus across its geodesic boundary. Its metric extends smoothly across the boundary and gives a closed hyperbolic surface of genus two. Developing the second half by reflection in ℓ gives its two holonomy generators CBC^{-1} and CAC^{-1}; the reversal of their order makes the boundary commutator K^{-1}. Consequently the holonomy of the double is exactly the projectivization of (2.1) at 1, with the prescribed presentation. The holonomy of a closed hyperbolic surface is discrete and faithful. □

By openness of the quasifuchsian locus, there is an ordinary complex neighborhood $V \subset D$ of 1 such that ρ_w is projectively faithful for every $w \in V$ [8, Section 4.3].

Lemma 2.2 (Galois approximation on the rational curve)**.** *The set of parameters $z \in D \cap \mathbb{Q}(i, \sqrt{2})$ such that ρ_z is projectively faithful and*

$$\operatorname{Im}\left((\operatorname{tr} A(z))^2\right) \neq 0 \tag{2.2}$$

is dense in D.

Proof. Let σ be the automorphism of $\mathbb{Q}(i, \sqrt{2})$ fixing i and sending $\sqrt{2}$ to $-\sqrt{2}$. The map

$$\mathbb{Q}(i)^2 \longrightarrow \mathbb{C}^2, \qquad (r, s) \longmapsto (r + s\sqrt{2}, r - s\sqrt{2})$$

has dense image, since it is the restriction of an invertible complex-linear map and $\mathbb{Q}(i)$ is dense in $\mathbb{C}$. Thus, for any nonempty open $U \subset D$, we can choose z with $z \in U$ and $\sigma(z) \in V$ simultaneously. All matrix entries in (2.1) are rational functions over $\mathbb{Q}$, so

$$\sigma(\rho_z(\gamma)) = \rho_{\sigma(z)}(\gamma) \qquad (\gamma \in \Gamma).$$

If $\rho_z(\gamma) = \pm I$, applying σ gives the same equality at $\sigma(z)$. Projective faithfulness there implies $\gamma = e$. This proves projective faithfulness at z and, in particular, ordinary faithfulness into $\mathrm{SL}_2(\mathbb{C})$.

Finally

$$\operatorname{tr} A(z) = \frac{2 + 4z + 6z^2}{1 + 2z}$$

is nonconstant holomorphic on D. The square of this function is also nonconstant. By the open mapping theorem, its real locus has empty interior. We may therefore first shrink U to a nonempty open subset satisfying (2.2), and then make the same simultaneous approximation. □

For each parameter from Lemma 2.2, the image group has no finite orbit on $\mathbb{CP}^1$: a subgroup with a finite orbit is either finite or virtually solvable, whereas Γ is neither. Condition (2.2) implies that $A(z)$ has eigenvalues of distinct absolute values and thus is proximal. It also excludes an invariant generalized circle. Indeed, after conjugating such a circle to $\mathbb{RP}^1$, matrices preserving it have real or purely imaginary traces, so all their squared traces are real. These checks verify precisely strong irreducibility, proximality, and the circle exclusion in Rapaport–Ren's theorem. Algebraic entries verify its Diophantine assumption [12, Lemma 6.1]; the same number-field norm argument applies to complex entries, as recorded after Definition 1.1 in [15].

3. Density of the full entropy spectrum

We first record the entropy identity used in the proof. Let π be a projectively faithful representation of a finitely supported measured group (G, ϑ) into $\mathrm{SL}_2(\mathbb{C})$, where $\operatorname{supp}\vartheta$ generates G as a semigroup. Suppose its image is strongly irreducible, proximal, and preserves no generalized circle. The unique stationary measure η on $\mathbb{CP}^1$ defines a genuine ϑ-boundary: random translates contract to point masses, and the limiting map factors through the Poisson boundary [7, Theorem 2.3]. This statement also applies to nondiscrete images. Write χ for the top Lyapunov exponent of $\pi_*\vartheta$.

The Ledrappier–Young identity in [15, Theorem 2.14] gives

$$h_\vartheta(\mathbb{CP}^1, \eta) = 2\chi \dim \eta. \tag{3.1}$$

To identify its entropy convention, let X_1 be the first increment and L the limiting projective point. The quantity $H(p) - \Delta$ in that theorem is

$$H(X_1) - H(X_1 \mid L) = I(X_1; L) = \sum_g \vartheta(g) D(\pi(g)_*\eta \| \eta) = h_\vartheta(\mathbb{CP}^1, \eta).$$

When the matrix entries are algebraic, the Diophantine property follows from the number-field norm estimate; see the discussion following [15, Definition 1.1] and [12, Lemma 6.1]. Faithfulness preserves the atom weights of every convolution power, so the matrix walk has asymptotic discrete entropy $h_\vartheta = h_{\mathrm{RW}}(G, \vartheta)$. The dimension formula [15, Theorem 1.2] then reads

$$\dim \eta = \min\{2, h_\vartheta/(2\chi)\}, \qquad h_\vartheta(\mathbb{CP}^1, \eta) = \min\{h_\vartheta, 4\chi\}. \tag{3.2}$$

Rapaport–Ren use base-two logarithms; changing both entropy and Lyapunov exponent to natural logarithms preserves these identities.

Proof of Theorem A: density and arithmetic control. Put $K_0 = \mathbb{Q}(i, \sqrt{2})$ and $h_\vartheta = h_{\mathrm{RW}}(\Gamma_g, \vartheta)$. Nonamenability of Γ_g and the entropy criterion imply $h_\vartheta > 0$ [14]. First take $g = 2$ and use the rational family ρ_z, $z \in D$, from (2.1). Let $\mathcal{G} \subset D \cap K_0$ be the dense set from Lemma 2.2. For every $z \in \mathcal{G}$, the representation is projectively faithful and its image is strongly irreducible, proximal, and preserves no generalized circle. This set of parameters depends on the representations, not on the probability weights or the support of ϑ.

We record why these properties remain available in higher genus. There is a subgroup $H_g \leq \Gamma_2$ of index $g-1$ isomorphic to Γ_g: for $g > 2$, take the kernel of an epimorphism $\Gamma_2 \to \mathbb{Z}/(g-1)\mathbb{Z}$, and use the Euler characteristic of its covering surface. For $g = 2$, set $H_2 = \Gamma_2$. Fix an identification $\Gamma_g \cong H_g$ and set $\pi_z = \rho_z|_{H_g}$. Each matrix entry of $\pi_z(\gamma)$ is a rational function over $\mathbb{Q}$. The representation π_0 is trivial, whereas π_1 is projectively faithful and cocompact Fuchsian. For $z \in \mathcal{G}$, projective faithfulness and algebraic entries are inherited by restriction.

The restricted image has no finite projective orbit: a subgroup of $\mathrm{PSL}_2(\mathbb{C})$ with a finite orbit is finite or virtually solvable, whereas H_g is neither. A positive power of the loxodromic matrix $A(z)$ belongs to $\rho_z(H_g)$, so the restricted image is proximal. To verify the circle condition, suppose that $\rho_z(H_g)$ preserves a generalized circle C. The normal core $N_g = \bigcap_{\gamma \in \Gamma_2} \gamma H_g \gamma^{-1}$ has finite index in Γ_2 and is contained in H_g. Its image is again non-elementary and contains loxodromic elements. Such a group can preserve at most one generalized circle: the fixed points of its loxodromic elements lie on every invariant circle, and non-elementarity provides at least three distinct such points. Normality therefore implies $\rho_z(\gamma)C = C$ for every $\gamma \in \Gamma_2$, since each $\rho_z(\gamma)C$ is another $\rho_z(N_g)$-invariant circle. This contradicts the circle exclusion for $\rho_z(\Gamma_2)$. Thus the restricted image satisfies all three dynamical hypotheses of Rapaport–Ren.

Now keep g and ϑ fixed, and write $\chi_\vartheta(z)$ for the top Lyapunov exponent of $\pi_{z*}\vartheta$. Enumerate the finite support of ϑ as $\{s_1, \dots, s_m\}$ and put $p_j = \vartheta(s_j) > 0$. The matrix tuple $(\pi_z(s_1), \dots, \pi_z(s_m))$ varies continuously in $\mathrm{GL}(2, \mathbb{C})^m$. Bocker-Neto and Viana's Theorem A [2] proves continuity of both extremal Lyapunov exponents for arbitrary complex two-dimensional matrix tuples with positive probability weights. It applies at every point of D, including reducible tuples and collisions of labeled matrix values. Consequently χ_ϑ is continuous on D, and $\chi_\vartheta(0) = 0$. No inversion symmetry is used here.

At $z = 1$, the orbit of the cocompact Fuchsian group $\pi_1(\Gamma_g)$ satisfies an upper counting bound Ce^R in $\mathbb{H}^2$. Thus, for every $q > 1$,

$$Z_q = \sum_{\gamma \in \Gamma_g} e^{-q d(o, \pi_1(\gamma)o)} < \infty.$$

The relative-entropy inequality against these normalized exponential weights gives, for the ϑ-random walk X_n,

$$H(\vartheta^{*n}) \leq q\, \mathbb{E} d(o, \pi_1(X_n)o) + \log Z_q.$$

The displacement drift equals $2\chi_\vartheta(1)$. Dividing by n and then letting $q \downarrow 1$ yields

$$h_\vartheta \leq 2\chi_\vartheta(1).$$

Since D is path connected, every $t \in [0, h_\vartheta]$ therefore has a parameter $z_t \in D$ with $4\chi_\vartheta(z_t) = t$.

For $z \in \mathcal{G}$, the support of $\pi_{z*}\vartheta$ generates $\pi_z(\Gamma_g)$ as a semigroup. Its algebraic entries give the Diophantine property, as recorded after Definition 1.1 of [15]. Faithfulness preserves every atom weight of every convolution power, so its random-walk entropy is h_ϑ. The stationary measure $\eta_{z,\vartheta}$ on $\mathbb{CP}^1$ is a genuine ϑ-boundary by [7, Theorem 2.3]; the equivalence in [15, Section 2.9] gives the required real Zariski density. Applying Theorems 1.2 and 2.14 of [15], with the first-step conditional-entropy identification above, gives

$$h_\vartheta(\mathbb{CP}^1, \eta_{z,\vartheta}) = \min\{h_\vartheta, 4\chi_\vartheta(z)\}.$$

Approximate z_t by parameters in $\mathcal{G}$ and use continuity of χ_ϑ to obtain entropies converging to t. All boundary entropies are bounded by h_ϑ, proving equality of closures. Finally, $\eta_{z,\vartheta}$ is nonatomic, and each nonidentity group element fixes at most two projective points. Projective faithfulness and countability of Γ_g imply essential freeness. □

The following refinement completes the last assertion of Theorem A.

Corollary 3.1. *The approximating boundaries constructed above may all be chosen so that the representation has dense image in* $\mathrm{SL}_2(\mathbb{C})$ *and the stationary measure has full support and exact dimension two on* $\mathbb{CP}^1$.

Proof. Write $h_\vartheta = h_{\mathrm{RW}}(\Gamma_g, \vartheta) > 0$. It suffices to approximate targets $t \in (0, h_\vartheta)$, since the endpoints are limits of interior targets. In the preceding construction, choose the good parameters sufficiently close to z_t that $4\chi_\vartheta(z) < h_\vartheta$. Rapaport–Ren's dimension formula then gives exact dimension two. If the projective image were discrete, packing balls in $\mathbb{H}^3$ would give an orbit bound Ce^{2R}. The preceding relative-entropy argument, now with $q > 2$, would imply $h_\vartheta \le 4\chi_\vartheta(z)$, a contradiction.

Let L be the closure of $\pi_z(\Gamma_g)$ in $\mathrm{SL}_2(\mathbb{C})$. Its Lie algebra $\mathfrak{l}$ is nonzero because L is not discrete. The condition that $\mathrm{Ad}(g)$ preserves the real vector subspace $\mathfrak{l}$ is a real algebraic condition on g. Since $\pi_z(\Gamma_g)$ is real Zariski dense, $\mathfrak{l}$ is invariant under all of $\mathrm{Ad}(\mathrm{SL}_2(\mathbb{C}))$. It is therefore a nonzero ideal in the simple real Lie algebra $\mathfrak{sl}_2(\mathbb{C})$, and must equal that Lie algebra. Hence L is open; connectedness gives $L = \mathrm{SL}_2(\mathbb{C})$.

Finally, stationarity and semigroup generation make the closed support of $\eta_{z,\vartheta}$ invariant under $\pi_z(\Gamma_g)$, and then under its closure. Transitivity of $\mathrm{SL}_2(\mathbb{C})$ on $\mathbb{CP}^1$ implies that the support is the entire projective line. □

Remark 3.2. Exact dimension two does not imply absolute continuity with respect to spherical area. No absolute-continuity assertion is needed for the density theorem or its corollary.

3.1. **The fixed genus-two walk.** For the rest of the paper, unless another measure is specified, use (Γ, μ) from (1.1). Let $\chi(z)$ denote the top Lyapunov exponent of the eight labeled matrices $\rho_z(s)$, each with weight $1/8$. Matrix collisions do not change this convention. The proof above gives continuity of χ on D, with $\chi(0) = 0$ and $h \le 2\chi(1)$. Write $Y_z = (\mathbb{CP}^1, \eta_z)$ for the unique projective stationary boundary whenever the image is strongly irreducible and proximal. At the dense algebraic faithful parameters from Lemma 2.2,

$$h_\mu(Y_z) = \min\{h, 4\chi(z)\}. \tag{3.3}$$

This parameter set is countable. The connected parameter domain is used to interpolate χ, not to interpolate boundary entropy.

Corollary 3.3. *For the fixed walk,* $\overline{\mathscr{H}_\partial(\Gamma, \mu)} = [0, h]$, *with approximating boundaries having all the properties in Theorem A.*

Proof. Apply Theorem A with $g = 2$ and $\vartheta = \mu$. □

4. Geometry and separation of Lyapunov levels

Let

$$Q = \{z \in D : \Re z > 0,\ \Im z > 0\}, \qquad N = \{z \in Q : \rho_z \text{ is not projectively faithful}\},$$

where D is the domain of the rational family. Throughout this section, $h = h_{\mathrm{RW}}(\Gamma, \mu)$ and χ denotes its Lyapunov exponent.

Proposition 4.1 (Faithful representations on exact Lyapunov levels)**.** *Every representation in Q is non-elementary, proximal, and preserves no generalized circle. The set N is countable. For every $0 < c < \chi(1)$, the level*

$$L_c = \{z \in Q : \chi(z) = c\}$$

satisfies $\dim_H(L_c \setminus N) \geq 1$*. In particular, for every $0 < t < h$, the family contains a projectively faithful, real-Zariski-dense representation with $\chi = t/4$.*

Proof. For each nonidentity $\gamma \in \Gamma$, the equation $\rho_z(\gamma) = \pm I$ has finitely many solutions in D: its entries are rational functions, and neither identity holds identically because ρ_1 is projectively faithful. Taking the union over γ proves countability of N.

Write

$$x(z) = \operatorname{tr} A(z) = \frac{1}{2} + 3z + \frac{3}{2(1+2z)}.$$

For $z = u + iv \in Q$,

$$\Re x(z) = \frac{1}{2} + 3u + \frac{3(1+2u)}{2|1+2z|^2} > 0, \qquad \Im x(z) = 3v\frac{|1+2z|^2 - 1}{|1+2z|^2} > 0.$$

Thus $x(z)^2$ is nonreal. Consequently $A(z)$ is loxodromic, and the image preserves no generalized circle. Moreover,

$$\operatorname{tr}[A(z), B(z)] = 2 - \frac{32z^4(z+2)^2}{(1+2z)^2} \neq 2 \qquad (z \in Q),$$

so the two-generator reducibility criterion excludes a common fixed point. If A and B preserved an unordered pair, each would fix both points: an element interchanging the pair has trace zero, whereas $\operatorname{tr} A = \operatorname{tr} B = x \neq 0$. This again contradicts irreducibility. A finite invariant set of at least three points would give a finite projective image, contradicting loxodromicity. Hence the family is non-elementary on Q. Its real-Zariski density follows from [15, Lemma 2.11].

The set Q is a quadrant with finitely many points removed and is connected. Continuity of χ and the limits at 0 and 1 give points $p_-, p_+ \in Q$ with $\chi(p_-) < c < \chi(p_+)$. A smooth embedded arc joining these points has a sufficiently narrow tubular neighborhood, giving a bi-Lipschitz embedding

$$F : [0,1] \times [-\varepsilon, \varepsilon] \longrightarrow Q$$

with $\chi(F(0,s)) < c < \chi(F(1,s))$ for every s. Each arc $F([0,1] \times \{s\})$ meets L_c. Therefore the Lipschitz map obtained from F^{-1} by projection onto its second coordinate maps $L_c \cap \operatorname{im} F$ onto $[-\varepsilon, \varepsilon]$. It follows that $\dim_H L_c \geq 1$. Removing the countable set N preserves this lower bound. Finally, $h \leq 2\chi(1)$ implies $t/4 < \chi(1)$ for $0 < t < h$. □

Open problem: separation on individual levels. Let $E \subset Q$ consist of the parameters for which $\rho_z(S)$ fails the weak Diophantine condition of [15, Definition 1.1]. Does

$$L_{t/4} \setminus (N \cup E) \neq \varnothing \qquad \text{for every } 0 < t < h? \tag{4.1}$$

An affirmative answer, together with the dimension formula, would prove full entropy realization. The estimate $\dim_H E < 1$ would suffice by Proposition 4.1. In contrast, density or full planar measure of $Q \setminus E$ does not imply (4.1), since a prescribed level may lie in an exceptional null set. The arithmetic construction supplies a countable dense set of parameters outside E.

4.1. **Parametric separation and symmetric generating sets.** Solomyak–Takahashi [16, Theorem 2.10] establishes a zero-dimensional exceptional parameter set for real-analytic families of contracting projective iterated function systems satisfying the infinite-word nondegeneracy condition of their Definition 2.9. Their Lemmas 2.13 and 2.15 obtain a uniform derivative order from compactness of the analytic coding maps and then deduce the dimension bound. Strict invariance of an open set supplies the required contraction.

Symmetric matrix sets do not satisfy this strict-invariance hypothesis [16, Remark 1.3(1)]. In addition, the strong Diophantine condition in that theorem separates distinct formal words, whereas symmetric surface-group tuples have cancellations and the defining relation at every parameter. Indexing matrices by distinct group elements removes those permanent equalities but leaves the contraction and coding hypotheses unverified. Rapaport–Ren [15, Section 1.2] treats parametric exceptional-set estimates as an open direction; its pointwise theorem supplies no such estimate for this family.

4.2. **Kernel parameters and order of contact.**

Proposition 4.2 (Unbounded order of contact with the identity)**.** *If $z_0 \in D$ is not projectively faithful, then for every integer $k \geq 0$ there is a nontrivial $w \in \Gamma$ such that*

$$\rho_z(w) - I = O((z - z_0)^{k+1}), \qquad \rho_z(w) \not\equiv I.$$

At $z_0 = 0$, iterated commutators of the first two standard generators have this property.

Proof. The nontrivial normal subgroup $\ker \bar{\rho}_{z_0}$ contains a free subgroup of rank two. Indeed, in a faithful Fuchsian realization, conjugate a nontrivial kernel element by an element moving its fixed point pair. Suitable powers of these independent hyperbolic elements generate a free subgroup by ping-pong. Taking further squares gives free generators u, v with $\rho_{z_0}(u) = \rho_{z_0}(v) = I$. Set $w_1 = [u, v]$ and $w_{j+1} = [u, w_j]$. These words are nontrivial. The elementary estimate

$$U(z) = I + O((z - z_0)^a), \quad V(z) = I + O((z - z_0)^b) \implies [U(z), V(z)] = I + O((z - z_0)^{a+b})$$

gives $\rho_z(w_j) - I = O((z - z_0)^{j+1})$. Projective faithfulness at 1 excludes an identically trivial word matrix. At zero one can take $u = a_1, v = b_1$. □

Proposition 4.3 (Density of kernel parameters)**.** *The nonfaithful set N is dense in*

$$U_{\text{low}} = \{z \in Q : \chi(z) < h/4\}.$$

On no nonempty open subset of U_{low} can a fixed finite number of derivatives separate every nontrivial word matrix from the identity with positive lower bounds uniform in the parameter.

Proof. The family on Q is non-elementary, generally faithful, and nontrivial: countability of N gives general faithfulness, and $\operatorname{tr}^2 A$ is nonconstant. The Sullivan–Bers stability theorem, in the form of [5, Theorem 2.6], implies that any open set consisting of faithful representations consists of discrete faithful representations.

For a discrete faithful representation into $\mathrm{PSL}_2(\mathbb{C})$, its orbit in $\mathbb{H}^3$ is uniformly separated. Ball packing gives

$$\#\{\gamma : d(o, \rho(\gamma)o) \leq R\} \leq Ce^{2R}.$$

The Gibbs entropy inequality with comparison weights proportional to $e^{-qd(o,\rho(\gamma)o)}$, $q > 2$, and the displacement drift 2χ imply $h \leq 2q\chi$. Letting $q \downarrow 2$ gives $h \leq 4\chi$. Thus no nonempty open subset of U_{low} can consist entirely of faithful representations, proving density of N.

Given such an open subset and an integer k, choose $z_0 \in N$ there. Proposition 4.2 supplies a nontrivial word for which

$$\partial_z^j(\rho_z(w) - I)\big|_{z=z_0} = 0 \qquad (0 \leq j \leq k).$$

This excludes even word-dependent positive lower bounds for the maximum of these derivatives, uniform in the parameter. Restricting the estimate to faithful parameters does not restore a positive lower bound: faithful parameters approach z_0, and the derivatives are continuous. □

Corollary 4.4 (Nonharmonicity in the subcritical region)**.** *The function χ is harmonic on no nonempty open subset of U_{low}. In particular it is nowhere locally constant there.*

Proof. Deroin–Dujardin [5, Definition 3.2 and Theorem 3.9] identifies the support of $dd^c\chi$ with the bifurcation locus. The admissibility conditions in their Section 3 hold here: the family is holomorphic, generally faithful, nontrivial, and everywhere non-elementary. The fixed finitely supported generating measure satisfies their exponential moment condition (8). The stability locus is the interior of the discrete-faithful locus [5, Theorem 2.6], so the packing inequality in the preceding proof excludes every stable point of U_{low}. Hence $U_{\mathrm{low}} \subset \mathrm{supp}(dd^c\chi)$, which excludes harmonicity on any nonempty open subset. □

The derivative obstruction concerns estimates uniform on complex open sets. Equation (4.1) asks instead for one weakly Diophantine parameter on each prescribed level. Density of kernel parameters in the ambient domain does not resolve that question.

5. Generic exact realization

Lower semicontinuity of entropy and the absence of constant Lyapunov patches give exact realization for a comeager set of target values.

Lemma 5.1 (Generic fibers of a continuous scalar map)**.** *Let $X \subset \mathbb{R}^2$ be open, let I be an open interval, and let $p : X \to I$ be a continuous surjection which is constant on no nonempty open subset. If $G \subset X$ is dense G_δ, there is a dense G_δ set $R \subset I$ such that, for each $t \in R$, $G \cap p^{-1}(t)$ is a dense G_δ subset of the nonempty perfect Polish space $p^{-1}(t)$. In particular this intersection has cardinality continuum.*

Proof. Let E consist of the values attained at local extrema of p. A countable base shows that E is countable: every such value is the supremum or infimum of p on a basis element. Since every level is closed and nowhere dense, $X' = X \setminus p^{-1}(E)$ is dense G_δ, and $I' = I \setminus E$ is dense G_δ. The restricted map $p : X' \to I'$ is a continuous open surjection. Indeed a connected ball about a point of X' contains values on both sides of its value, and its image therefore contains an interval about that value. Deleting E preserves this neighborhood statement in the relative topology.

Write $G \cap X' = \bigcap_n U_n$ with each U_n open dense in X', and take a countable open base (V_j) for X'. For each n, j, $p(V_j \cap U_n)$ is open dense in the open set $p(V_j)$. To check density, any nonempty relatively open set $W \subset p(V_j)$ has a nonempty open inverse image in V_j, which meets U_n. It follows that $p(V_j) \setminus p(V_j \cap U_n)$ is nowhere dense in I': its closure has empty interior in $p(V_j)$ and can add only points on the boundary of that open set. Remove the countable union of these closures. The remaining set R is dense G_δ in I' and hence in I. For $t \in R$, every U_n meets every nonempty basic open subset of $p^{-1}(t)$, so it is dense in that fiber. The fiber is closed in the Polish space X', and is therefore Baire; this proves the dense G_δ conclusion.

Finally, the fiber has no isolated points. Otherwise a sufficiently small punctured disk about an isolated fiber point would be connected and contain no other point of the same level. The continuous function $p - t$ would have constant sign there, making its center a local extremum, contrary to $t \notin E$. A nonempty perfect Polish space and each of its dense G_δ subsets have cardinality continuum. □

Lemma 5.2 (Entropy saturation is residual in parameter space)**.** *For the surface-group family, put*

$$X = \{z \in Q : 0 < 4\chi(z) < h\}, \qquad e(z) = h_\mu(\mathbb{CP}^1, \eta_z),$$

where η_z is the unique stationary measure. There is a dense G_δ subset $G \subset X$ on which ρ_z is projectively faithful and $e(z) = 4\chi(z)$.

Proof. Non-elementarity throughout Q gives weak continuity of η_z: any weak limit along convergent parameters is stationary for the limit tuple and hence equals its unique stationary measure. Write D_{KL} for relative entropy of probability measures. The change-of-variables form of Furstenberg entropy is

$$e(z) = \frac{1}{8} \sum_{s \in S} D_{\mathrm{KL}}\big(\rho_z(s)_* \eta_z \,\|\, \eta_z\big).$$

Each summand is lower semicontinuous in z. One can see this directly from the variational identity, for probability measures α, β on the compact space $\mathbb{CP}^1$,

$$D_{\mathrm{KL}}(\alpha \| \beta) = \sup_{f \in C(\mathbb{CP}^1)} \left(\int f \, d\alpha - \log \int e^f \, d\beta \right).$$

Thus e is lower semicontinuous. The Ledrappier–Young identity $e(z) = 2\chi(z) \dim \eta_z$ gives $e(z) \le 4\chi(z)$ without any separation assumption. The dense faithful algebraic parameters from Lemma 2.2 satisfy equality on X, by (3.3).

It follows that, for every $n \ge 1$,

$$O_n = \{z \in X : e(z) > 4\chi(z) - 1/n\}$$

is open and dense. Intersect these sets and remove the countable nonfaithful set N. The result is a dense G_δ set G with the asserted properties. Notice that this argument does not assert weak Diophantinity on G; it proves the entropy conclusion directly. □

Theorem 5.3 (Generic exact realization throughout the full range)**.** *There is a dense G_δ set $R_\Gamma \subset (0, h)$ such that every $t \in R_\Gamma$ is the entropy of an essentially free boundary of the fixed (Γ, μ). Consequently*

$$(0, h) \setminus \mathscr{H}_\partial(\Gamma, \mu) \quad \textit{is meager.}$$

Proof. Use X and G from Lemma 5.2, and set $p(z) = 4\chi(z)$. Proposition 4.1 shows $p(X) = (0, h)$. Corollary 4.4, based on the support theorem for the bifurcation current, verifies the remaining hypothesis of Lemma 5.1. For t in its dense G_δ set, select $z \in G$ with $p(z) = t$. Then $e(z) = t$. The stationary projective space is a genuine boundary, as before. Its measure is nonatomic, and projective faithfulness implies essential freeness because a nonidentity Möbius transformation fixes at most two points. □

The exceptional target set is meager. This gives no assertion about its Lebesgue measure, and a particular prescribed value can still belong to it. A second construction will turn the corresponding generic statement for a free group into a complete numerical interval for the surface group.

6. Generic realization for the free group

Write $F = \langle x, y \rangle$, let

$$\sigma = \tfrac{1}{4}(\delta_x + \delta_{x^{-1}} + \delta_y + \delta_{y^{-1}}), \qquad h_2 = h_{\mathrm{RW}}(F, \sigma) = \tfrac{1}{2} \log 3.$$

Proposition 6.1. *There is a comeager set $R \subset (0, h_2)$ such that every $r \in R$ is the entropy of a genuine σ-boundary of F. These boundaries can be chosen to be essentially free projective boundaries of faithful representations $F \longrightarrow \mathrm{SL}_2(\mathbb{C})$.*

Proof. Use the two matrices $A(z), B(z)$ already defined in the explicit curve, now as the images of the two free generators. Denote this representation by π_z and its top Lyapunov exponent for the four equally weighted matrices by $\chi_F(z)$. This is a separate exponent from the eight-step surface-group exponent.

At $z = 0$ the representation is trivial. At $z = 1$ it is the faithful convex cocompact one-holed-torus representation checked above. The faithful neighborhood V for the surface-group family also gives faithfulness for this free subgroup throughout V. The same $\mathbb{Q}(i, \sqrt{2})$ Galois approximation proves that faithful algebraic parameters are dense. Every transcendental parameter is faithful as well: a nontrivial word equal to $\pm I$ there would make the corresponding rational-function identity hold identically, contradicting projective faithfulness at 1.

Use the same punctured first quadrant Q as in Proposition 4.1. Its trace and commutator calculations involve only A, B, so they prove that π_z is strongly irreducible, proximal, and preserves no generalized circle throughout Q.

The function χ_F is continuous. Its values approach zero as $z \to 0$ in Q, whereas $\chi_F(1) \geq h_2/2$ by the same hyperbolic-plane orbit-counting argument used at the surface-group endpoint. Since a quadrant minus finitely many points is path connected, every value in $(0, h_2/4)$ occurs. Let

$$X = \{z \in Q : 0 < 4\chi_F(z) < h_2\}.$$

Then X is open and the range of $4\chi_F$ on X is $(0, h_2)$.

We check the absence of locally constant exponent patches, which is essential for selection of entropy values rather than just parameters. On every connected open subset of X, the family π_z is an admissible holomorphic family in Deroin–Dujardin's terminology: it is nonelementary throughout, generally faithful, and nontrivial because the squared trace $(\operatorname{tr} A(z))^2$ is nonconstant. The finitely supported generating measure σ satisfies all their moment conditions. By [5, Theorems 2.6 and 3.9], the support of $dd^c\chi_F$ is the bifurcation locus, and a stable open patch would consist of discrete faithful representations. For any such representation, packing in $\mathbb{H}^3$ gives

$$h_2 \leq 4\chi_F.$$

Indeed its discrete faithful orbit has at most Ce^{2R} points in an R-ball; the relative-entropy comparison with the weights $e^{-qd(o,go)}$, $q > 2$, gives $h_2 \leq 2q\chi_F$, and then let $q \downarrow 2$. This contradicts the definition of X. Thus χ_F is not harmonic, and in particular not constant, on any nonempty open subset of X.

For $z \in X$ let $e_F(z)$ be the Furstenberg entropy of the unique stationary probability on $\mathbb{CP}^1$. The stationary measures vary weakly continuously. The relative-entropy formula makes e_F lower semicontinuous, and the dimension formula gives $e_F(z) \leq 4\chi_F(z)$. At the dense faithful algebraic parameters, Rapaport–Ren gives equality, since $4\chi_F < h_2$. It follows that

$$G = \{z \in X : e_F(z) = 4\chi_F(z)\}$$

is a dense G_δ subset of X. Intersecting it with the comeager set of faithful parameters preserves this property. Lemma 5.1, applied to the continuous nowhere locally constant function $4\chi_F : X \to (0, h_2)$ and this comeager set, gives a comeager set R of realized values.

The projective stationary spaces are genuine boundaries because strong irreducibility and proximality give almost-sure convergence of the random translates of their stationary measures to point masses. Their measures are nonatomic. Faithfulness then makes the F-actions essentially free, since each nonidentity projective transformation fixes at most two points. □

7. Boundary factors and entropy limits

The results in this and the next two sections apply to any countable group G equipped with a finitely supported probability measure μ whose support generates G as a semigroup. Let (B, ν)

be its Poisson boundary and put $H = h_\mu(B, \nu)$. We identify boundary factors with complete G-invariant sub-σ-algebras of B, modulo null sets. For such an algebra A, write P_A for conditional expectation on $L^2(B, \nu)$ and $h(A)$ for its Furstenberg entropy. Joins and meets are taken in this lattice.

7.1. **Conditional derivatives and monotone limits.** For $g \in G$, put $r_g = d(g_*\nu)/d\nu$. If $\mu^{*k}(g) > 0$, stationarity gives

$$r_g \le \frac{1}{\mu^{*k}(g)}.$$

An analogous bound for g^{-1}, together with the Radon–Nikodym cocycle identity, gives a strictly positive lower bound for r_g. Semigroup generation ensures that both bounds exist for every g. In particular there are constants $0 < c \le C < \infty$ such that $c \le r_s \le C$ for every $s \in \operatorname{supp} \mu$.

The derivative on a factor pulls back to $P_A r_g$. With $\phi(u) = u \log u$, the change-of-variables form of entropy is therefore

$$h(A) = \sum_s \mu(s) \int_B \phi(P_A r_s)\, d\nu. \tag{7.1}$$

Consequently, strong convergence $P_{A_n} \to P_A$ implies $h(A_n) \to h(A)$, since ϕ is Lipschitz on $[c, C]$. For the uniform eight-generator surface walk one can take $c = 1/8$ and $C = 8$; symmetry also gives the equivalent formula

$$h(A) = -\frac{1}{8} \sum_{s \in S} \int_B \log(P_A r_s)\, d\nu.$$

Martingale convergence now proves continuity of entropy along increasing joins and decreasing meets:

$$A_n \uparrow A \implies h(A_n) \uparrow h(A), \qquad A_n \downarrow A \implies h(A_n) \downarrow h(A).$$

These are the monotone continuity statements in [6, Theorem C]. They concern factors of a fixed Poisson boundary and do not assert entropy continuity for arbitrary convergent representations. Theorem 11.1 exhibits precisely that distinction.

7.2. **Compactness and scalar closedness.** Scalar density alone does not provide a convergent sequence of factors. The compactness argument in [6, Section 4, Theorem D] invokes compactness of the space of all sub-σ-algebras in the strong conditional-expectation topology, citing Beissner–Tölle. That source was withdrawn after a gap was identified in its compactness proof [1].

The general compactness assertion fails. On a standard product probability space, let $X, Z_1, Z_2, \dots$ be independent signs, with X fair and $\mathbb{E}Z_n = a \in (0, 1)$. For $\mathcal{A}_n = \sigma(XZ_n)$ one has

$$\mathbb{E}[X \mid \mathcal{A}_n] = aXZ_n, \qquad \big\|\mathbb{E}[X \mid \mathcal{A}_n] - \mathbb{E}[X \mid \mathcal{A}_m]\big\|_2^2 = 2a^2(1 - a^2) \quad (m \ne n).$$

Thus the corresponding conditional expectations have no strongly convergent subsequence. This example concerns arbitrary subalgebras; it does not disprove compactness for invariant subalgebras of a particular Poisson boundary. It does show that the general compactness argument cannot establish that specialized conclusion.

Burton–Lupini–Tamuz likewise withdrew the compactness claim in Theorem 4.6 of the first version of their paper after finding a gap in its proof [4]. Their retained results on weak equivalence and entropy continuity do not establish compactness of the boundary entropy spectrum. None of these compactness claims is used below.

8. Submodularity and a complete entropy metric

Ordinary join-subadditivity appears in [17, Section 4, proof of Theorem 1.3]. We give a relative form with a common-factor correction. The proof uses nonstationary entropy; the initial relatively independent joining need not be stationary, and the limiting canonical joining need not be absolutely continuous with respect to it.

8.1. **Nonstationary entropy and relative joins.** For a probability measure m on a compact metrizable G-space define

$$\Phi_\mu(m) = \sum_g \mu(g) D(g_* m \| \mu * m).$$

This is the nonstationary entropy of [10, Definition 2.3]. It is weakly lower semicontinuous, satisfies $0 \le \Phi_\mu(m) \le H(\mu)$, and agrees with Furstenberg entropy when m is stationary. The mixture identity

$$\sum_i p_i D(m_i \| q) = \sum_i p_i D(m_i \| \overline{m}) + D(\overline{m} \| q), \qquad \overline{m} = \sum_i p_i m_i,$$

implies the nonstationary convolution identity

$$\Phi_{\mu*\kappa}(m) = \Phi_\mu(\kappa * m) + \Phi_\kappa(m). \tag{8.1}$$

These identities are [10, Lemma 2.8 and Proposition 2.10]. Their proofs use only mixtures and invertibility, so apply to the compact models used here. Indeed, condition on g in the average of $D(g_* h_* m \| \mu * \kappa * m)$ and use the intermediate mixture $g_*(\kappa * m)$. The within-mixture term is $\Phi_\kappa(m)$ by invariance of relative entropy under g; the remaining term is $\Phi_\mu(\kappa * m)$. In particular,

$$\Phi_{\mu^{*n}}(m) = \sum_{k=0}^{n-1} \Phi_\mu(\mu^{*k} * m), \qquad h_{\mu^{*n}}(A) = n h(A).$$

Proposition 8.1 (Submodularity of boundary entropy)**.** *If A, C are μ-boundaries and D is a common boundary factor, then*

$$h(A \vee C) + h(D) \le h(A) + h(C).$$

In particular, this holds for $D = A \wedge C$.

Proof. Choose simultaneous compact metrizable models in which the actions and the maps $A \to D$ and $C \to D$ are continuous. Such models can be obtained by taking countable families of bounded Borel functions separating points, closing these families under group translates and pullbacks from D, and embedding the spaces by their coordinates in countable products of compact intervals. The closures carry continuous actions and factor maps; the measures are concentrated on the original Borel images. Measurable actions may first be restricted to invariant conull sets so that the group laws hold everywhere.

Disintegrate the stationary measures over D and set

$$m_0 = \int_D \nu_d^A \otimes \nu_d^C \, d\nu^D(d), \qquad m_n = \mu^{*n} * m_0.$$

These are measures on the compact fiber product $A \times_D C$. Writing $r_E(g, \cdot) = d(g_* \nu^E)/d\nu^E$ for $E = A, C, D$, the conditional derivative in each fiber is the total-space derivative divided by the base derivative. Hence

$$\frac{d(g_* m_0)}{dm_0}(a, c) = \frac{r_A(g, a) r_C(g, c)}{r_D(g, d)}, \qquad d = \pi_A(a) = \pi_C(c).$$

For each fixed word these derivatives are bounded above and away from zero, as in Section 7. All logarithms in the finite sums below are therefore integrable. Integrating against g_*m_0 and using its marginals gives

$$\sum_g \mu^{*n}(g) D(g_*m_0 \| m_0) = n\big(h(A) + h(C) - h(D)\big).$$

The mixture identity then yields

$$\begin{aligned} \Phi_{\mu^{*n}}(m_0) &= n\big(h(A) + h(C) - h(D)\big) - D(m_n \| m_0) \\ &\le n\big(h(A) + h(C) - h(D)\big). \end{aligned}$$

Along almost every random-walk path, the two marginals of the translate of m_0 converge to point masses, since A and C are boundaries. Every coupling of those marginals converges to the point mass on the corresponding pair. Taking expectations shows that m_n converges weakly to the canonical boundary joining λ. Thus

$$h(A \vee C) = \Phi_\mu(\lambda) \le \liminf_{k\to\infty} \Phi_\mu(m_k).$$

Cesaro averaging and (8.1) prove the assertion. □

8.2. **The entropy metric and summable perturbations.**

Proposition 8.2 (A complete metric on the boundary lattice)**.** *On the boundary lattice modulo null sets,*

$$d(A, C) = 2h(A \vee C) - h(A) - h(C)$$

is a complete metric. For every boundary Z,

$$d(A \vee Z, C \vee Z) \le d(A, C), \qquad |h(A) - h(C)| \le d(A, C).$$

Proof. Monotonicity gives nonnegativity. If $d(A, C) = 0$, then $h(A \vee C) = h(A) = h(C)$, and strict entropy monotonicity between boundary factors gives $A = C$. For completeness, strictness follows from Lemma 9.1 below: zero entropy deficit between nested factors makes their projections agree on the dense span of the derivatives, and hence on L^2. Submodularity over the common factor B gives

$$h(A \vee C) \le h(A \vee B \vee C) \le h(A \vee B) + h(B \vee C) - h(B),$$

which is the triangle inequality. Applying submodularity over A to $A \vee C$ and $A \vee Z$ gives

$$h(A \vee C \vee Z) - h(A \vee Z) \le h(A \vee C) - h(A).$$

Add the analogous inequality with C in place of A to obtain join contraction. The entropy estimate follows from $h(A \vee C) \ge \max\{h(A), h(C)\}$.

Suppose first that $\sum_n d(A_n, A_{n+1}) < \infty$. Put

$$T_N = \bigvee_{n \ge N} A_n, \qquad \epsilon_N = \sum_{n \ge N} \big(h(A_n \vee A_{n+1}) - h(A_n)\big).$$

These tails tend to zero. Submodularity bounds the entropy increment from adding A_{j+1} to $A_N \vee \cdots \vee A_j$ by $h(A_j \vee A_{j+1}) - h(A_j)$. Increasing-join continuity therefore gives

$$0 \le h(T_N) - h(A_N) \le \epsilon_N. \tag{8.2}$$

Let $A = \bigwedge_N T_N$. Decreasing-meet continuity implies $h(T_N) \to h(A)$, and $A \vee A_N \le T_N$ gives

$$d(A, A_N) \le h(T_N) - h(A) + h(T_N) - h(A_N) \longrightarrow 0.$$

Every Cauchy sequence has a subsequence with summable adjacent distances. The preceding argument gives its limit, and the triangle inequality gives the same limit for the original sequence. □

Corollary 8.3 (Exact entropy from summable join costs)**.** *For any sequence of boundaries (A_n), the following conditions are equivalent:*

$$\sum_n \big(h(A_n \vee A_{n+1}) - h(A_n)\big) < \infty, \qquad \sum_n d(A_n, A_{n+1}) < \infty.$$

Under either condition, A_n converges in d to the genuine boundary

$$A = \bigwedge_{N=1}^{\infty} \bigvee_{n \geq N} A_n.$$

In particular, if $h(A_n) \to t$, then $h(A) = t$.

Proof. Writing $c_n = h(A_n \vee A_{n+1}) - h(A_n)$, we have $0 \leq c_n \leq d(A_n, A_{n+1})$ and the telescoping identity

$$\sum_{n=1}^{N} d(A_n, A_{n+1}) = 2 \sum_{n=1}^{N} c_n + h(A_1) - h(A_{N+1}).$$

Nonnegativity of entropy proves equivalence. The tail-join argument above proves convergence to the displayed A, and entropy is 1-Lipschitz for d. □

Completeness does not imply compactness or closedness of the scalar entropy image. Further metric consequences are collected in Corollary 10.5.

9. Quantitative factor selection and relative extensions

9.1. **Entropy deficit and conditional expectations.** The next estimate relates the entropy metric to the realization of factors as conditional-expectation operators. It also gives explicit constants for the eight-generator surface walk.

Lemma 9.1 (Nested entropy deficit controls conditional derivatives)**.** *Suppose $A \subseteq C$, put $\Delta = h(C) - h(A)$, and choose $k \geq 1$ such that $\mu^{*k}(g) > 0$. Then*

$$\|(P_C - P_A) r_g\|_2 \leq \frac{\sqrt{2k}}{\mu^{*k}(g)} \sqrt{\Delta}.$$

For the uniform eight-generator walk, this implies

$$\|(P_C - P_A) r_g\|_2 \leq \sqrt{2\ell}\, 8^{\ell} \sqrt{\Delta} \qquad (|g| = \ell \geq 1).$$

Moreover, the linear span of $\{r_g : g \in G\}$ is dense in $L^2(B, \nu)$. For every $f \in L^2(B, \nu)$ there is consequently a modulus $\omega_f(\delta) \to 0$ as $\delta \downarrow 0$, independent of the nested pair, such that

$$\|(P_C - P_A) f\|_2 \leq \omega_f(\Delta).$$

Proof. Set $p = \mu^{*k}(g)$. Since $r_g \leq p^{-1}$, the function $\phi(u) = u \log u$ satisfies $\phi''(u) \geq p$ on the range of $u = P_A r_g$ and $v = P_C r_g$. Also $P_A v = u$, so the linear term in Taylor's formula integrates to zero. Hence

$$\int \big(\phi(v) - \phi(u)\big)\, d\nu \geq \frac{p}{2} \|v - u\|_2^2.$$

All the other terms in the entropy deficit for μ^{*k} are nonnegative by conditional Jensen. The g term alone therefore gives $k\Delta \geq (p^2/2)\|v - u\|_2^2$. For the uniform surface walk, $\mu^{*\ell}(g) \geq 8^{-\ell}$.

For density of the derivative span, suppose f is orthogonal to every r_g. Let β be the boundary map on the random-walk increment space, and let X_n be the product of the first n increments. The conditional law of β given those increments is $(X_n)_* \nu$, and thus

$$\mathbb{E}[f(\beta) \mid \omega_1, \ldots, \omega_n] = \int f r_{X_n}\, d\nu = 0.$$

Martingale convergence gives $f(\beta) = 0$ and hence $f = 0$. Finally, approximate f by a finite linear combination $F = \sum_j b_j r_{g_j}$. The difference of nested orthogonal projections has norm at most one, so the derivative estimates imply

$$\|(P_C - P_A)f\|_2 \le \|f - F\|_2 + \sum_j |b_j| \frac{\sqrt{2k_j}}{\mu^{*k_j}(g_j)} \sqrt{\Delta},$$

where $\mu^{*k_j}(g_j) > 0$; constant terms contribute zero. This proves the asserted uniform modulus. □

For arbitrary factors A, C, applying the lemma through $A \vee C$ and using $\sqrt{x} + \sqrt{y} \le \sqrt{2(x+y)}$ gives

$$\|(P_A - P_C)r_g\|_2 \le \frac{2\sqrt{k}}{\mu^{*k}(g)} \sqrt{d(A, C)}. \tag{9.1}$$

For the prescribed surface walk the right-hand constant is at most $2\sqrt{\ell}\, 8^\ell$. Density of the derivative span implies that d-convergence entails strong convergence of conditional expectations. No converse or continuity of the meet operation is asserted.

Proposition 9.2 (Selection by controlled canonical joins)**.** *Suppose $h(A_n) \to t$ and*

$$\lim_{N\to\infty} \sup_{m,n\ge N} h(A_m \vee A_n) = t.$$

Then A_n converges in d to a boundary A with $h(A) = t$, and $P_{A_n} \to P_A$ strongly. The same conclusions hold if $\sum_n d(A_n, A_{n+1}) < \infty$, equivalently if the one-sided join costs in Corollary 8.3 are summable.

Proof. The first hypothesis and $h(A_n) \to t$ imply $\sup_{m,n\ge N} d(A_m, A_n) \to 0$. Apply completeness and the 1-Lipschitz property of entropy, followed by (9.1). The summability assertion is Corollary 8.3. □

For comparison, the operator estimate alone already proves selection under the stronger condition $\sum_n \sqrt{d(A_n, A_{n+1})} < \infty$. It makes the projections strongly Cauchy on the derivative span and hence on L^2. A strong limit P of conditional expectations is self-adjoint, positive, unital and idempotent. If a bounded real f lies in its range, Jensen gives $P(f^2) \ge f^2$; equality of integrals gives equality. Polarization and bounded functional calculus show that the bounded range is an algebra. It is dense in the full range, by applying P to bounded truncations, so P is conditional expectation onto a complete subalgebra. Composition T_g by g is bounded on L^2 because r_g is bounded; passing to the limit in $(I - P_{A_n})T_g P_{A_n} = 0$ proves invariance. Formula (7.1) then proves entropy convergence. Submodularity replaces this square-root summability requirement by the equivalent summable-cost conditions above.

9.2. **Relative extension and exact realization.** An alternative to a Cauchy selection is to extend factors by arbitrarily small entropy increments. The relative entropy difference used below is consistent with the fiber-entropy decomposition in [11, eqs. (2.5)–(2.6)]; submodularity bounds the cost of adjoining a factor by its entropy over the common factor.

Proposition 9.3 (A relative extension criterion)**.** *For factors A, C, put $r_A(C) = h(C) - h(A \wedge C)$. Fix $T \in (0, H]$. If, for every A with $h(A) < T$, there are $C \not\le A$ with arbitrarily small $r_A(C)$, then*

$$[0, T] \subseteq \mathscr{H}_\partial(G, \mu).$$

More precisely, if $t \in (0, H]$ is not realized, there is a factor A with $a = h(A) < t$ such that

$$C \not\le A \quad \Longrightarrow \quad h(C) - h(A \wedge C) > t - a. \tag{9.2}$$

In particular, A contains every C with $r_A(C) \le t - a$, and therefore every C with $h(C) \le t - a$.

Proof. Let $\mathcal{P}_t$ be the factors of entropy at most t. For any nonempty chain $\mathcal{C} \subseteq \mathcal{P}_t$, put $s = \sup_{E\in\mathcal{C}} h(E)$ and choose an increasing sequence $E_n \in \mathcal{C}$ with $h(E_n) \to s$. Set $E = \bigvee_n E_n$. For each $F \in \mathcal{C}$, comparability gives

$$h(F \vee E_n) = \max\{h(F), h(E_n)\} \longrightarrow s.$$

Increasing-join continuity implies $h(F \vee E) = h(E) = s$, and strict entropy monotonicity gives $F \leq E$. Thus E is an upper bound in $\mathcal{P}_t$. The empty chain is bounded by the trivial factor, so Zorn's lemma gives a maximal $A \in \mathcal{P}_t$.

If t is not realized, then $a = h(A) < t$. For $C \not\leq A$, maximality implies $h(A \vee C) > t$. Submodularity yields

$$t < h(A \vee C) \leq a + h(C) - h(A \wedge C),$$

proving (9.2). Arbitrarily small relative costs contradict this inequality for every unrealized $t \leq T$. The trivial boundary realizes zero. □

The same Zorn argument proves full realization if every proper factor has strict extensions with arbitrarily small entropy increments. Proposition 9.3 gives a sufficient condition for constructing such extensions through canonical joins. In particular, taking $T = H/2$ reduces realization of the lower half-interval to small relative costs outside every A with $h(A) < H/2$. This allows candidates of substantial absolute entropy when most of that entropy already belongs to the common factor.

Neither scalar density nor the generic realization theorem verifies this extension condition. The small-extension theorem of Tamuz–Zheng [17, Theorem 1.3] applies to non-essentially-free free-group boundaries and does not provide it for arbitrary faithful surface-group boundaries. The selection and extension criteria above are sufficient conditions for further exact realization, not additional claims about the attained interval.

10. Applications

We apply generic realization first to products and then to the prescribed genus-two walk. The product theorem keeps the increment distribution explicit: each step acts on one coordinate, so boundary entropy is a weighted sum of the two coordinate entropies.

10.1. **Full realization for coordinate products.**

Theorem 10.1 (Exact realization for coordinate products)**.** *For $i = 1, 2$, let G_i be a countable group with a semigroup-generating probability measure μ_i of finite Shannon entropy, and suppose $0 < h_i := h_{\mathrm{RW}}(G_i, \mu_i) < \infty$. Assume that the boundary entropy spectrum of (G_i, μ_i) contains a comeager subset $R_i \subset (0, h_i)$. For $0 < p < 1$, define*

$$\lambda_p = p(\mu_1 \otimes \delta_e) + (1-p)(\delta_e \otimes \mu_2) \quad \text{on } G_1 \times G_2.$$

Then

$$h_{\mathrm{RW}}(G_1 \times G_2, \lambda_p) = ph_1 + (1-p)h_2, \qquad \mathscr{H}_\partial(G_1 \times G_2, \lambda_p) = [0, ph_1 + (1-p)h_2].$$

If the values in R_i are realized by essentially free boundaries, then every value in the open interval is realized by an essentially free boundary of $G_1 \times G_2$.

Proof. Let (B_i, η_i) be a μ_i-boundary. The product measure $\eta_1 \otimes \eta_2$ is λ_p-stationary. Realize the walk by independent coordinate choices, with probabilities $p, 1-p$, and independent increment streams of laws μ_1, μ_2. Both coordinates are updated infinitely often almost surely. Along the resulting sample path, the translates of each η_i converge to a point mass. Hence the translates of

the product measure also converge to a point mass. This proves that the stationary product is a genuine boundary. The entropy formula is

$$h_{\lambda_p}(B_1 \times B_2, \eta_1 \otimes \eta_2) = p h_{\mu_1}(B_1, \eta_1) + (1-p) h_{\mu_2}(B_2, \eta_2), \tag{10.1}$$

since the Radon–Nikodym derivative of a step is the derivative on its active coordinate. The product action is essentially free if both coordinate actions are essentially free.

For completeness, the random-walk entropy has the same weighted formula. Let N_n be the number of updates of the first coordinate by time n, and let Z_n be the position of the product walk. Conditional on $N_n = k$, its two coordinates are independent with laws μ_1^{*k} and $\mu_2^{*(n-k)}$. Therefore

$$H(Z_n \mid N_n) = \mathbb{E}\big[H(\mu_1^{*N_n}) + H(\mu_2^{*(n-N_n)})\big], \qquad 0 \le H(Z_n) - H(Z_n \mid N_n) \le \log(n+1).$$

The law of large numbers, the definition of h_i, and the bounds $H(\mu_i^{*k}) \le kH(\mu_i)$ show that the conditional entropy divided by n converges to $ph_1 + (1-p)h_2$. Thus this is the random-walk entropy and bounds every boundary entropy from above.

Fix $0 < t < ph_1 + (1-p)h_2$. The interval

$$J_t = \left(\max\left\{0, \frac{t-(1-p)h_2}{p}\right\}, \min\left\{h_1, \frac{t}{p}\right\} \right)$$

is nonempty. On J_t, both R_1 and the inverse image of R_2 under $r \mapsto (t-pr)/(1-p)$ are comeager. Their intersection is nonempty by the Baire category theorem. Selecting r_1 there and setting $r_2 = (t-pr_1)/(1-p)$ gives $t = pr_1 + (1-p)r_2$. Choose the corresponding boundaries and use (10.1). The trivial boundaries realize zero, and the two Poisson boundaries realize the upper endpoint. □

Corollary 10.2 (Full realization on products of surface groups)**.** *For the fixed* (Γ, μ) *and any* $0 < p < 1$,

$$\mathscr{H}_\partial(\Gamma \times \Gamma, p(\mu \otimes \delta_e) + (1-p)(\delta_e \otimes \mu)) = [0, h].$$

Every $t \in (0, h)$ *is realized by an essentially free boundary. Likewise, the coordinate product of two simple random walks on* F_2 *has boundary entropy spectrum* $[0, \frac{1}{2}\log 3]$.

Proof. Apply Theorem 10.1 to Theorem 5.3 and Proposition 6.1, respectively. □

10.2. **An explicit interval for the genus-two walk.**

Theorem 10.3. *For the walk* (1.1)*,*

$$[0, \tfrac{1}{2}\log 3] \subseteq \mathscr{H}_\partial(\Gamma, \mu).$$

Consequently $[0, c_* h] \subseteq \mathscr{H}_\partial(\Gamma, \mu)$*, where* $c_* = 2\log 3/(3\log 7)$*. The boundaries in this interval construction factor through* $F_2 \times F_2$ *and are not essentially free as* Γ*-spaces.*

Proof. With $F = \langle x, y \rangle$ and its simple random walk σ, define

$$q : \Gamma \longrightarrow F \times F, \qquad a_1 \mapsto (x, e), \quad a_2 \mapsto (y, e), \quad b_1 \mapsto (e, x), \quad b_2 \mapsto (e, y).$$

Both commutators in the defining relation map to the identity, and q is surjective. The push-forward measure is exactly

$$q_*\mu = \tfrac{1}{2}(\sigma \otimes \delta_e) + \tfrac{1}{2}(\delta_e \otimes \sigma).$$

Proposition 6.1 and Theorem 10.1 give every entropy in $[0, h_2]$, where $h_2 = \frac{1}{2}\log 3$. Pulling a boundary back along q preserves stationarity, the point-mass contraction property, and entropy. In particular,

$$h_\mu(B_1 \times B_2, \eta_1 \otimes \eta_2) = \tfrac{1}{2} h_\sigma(B_1, \eta_1) + \tfrac{1}{2} h_\sigma(B_2, \eta_2). \tag{10.2}$$

This proves the interval assertion for the prescribed measure. The estimate $c_* h \le \frac{1}{2}\log 3$ follows from (1.3). Finally $[a_1, b_1]$ is nontrivial in Γ and lies in $\ker q$, so all these pulled-back actions have a nontrivial kernel. □

The maximal entropy of this quotient walk is h_2. Thus this particular construction cannot yield an entropy larger than $\frac{1}{2}\log 3$; it imposes no upper bound on other boundary constructions for Γ.

10.3. **Abundance of proper boundaries.**

Corollary 10.4 (Abundance of proper, essentially free boundaries)**.** *Every nonempty open interval $J \subset (0, h)$ contains the entropy values of continuum many pairwise nonisomorphic, essentially free proper μ-boundaries of Γ. They may all be realized on $\mathbb{CP}^1$ with nonatomic, exact-dimensional stationary measures of dimension 2.*

Proof. Let R_Γ be the dense G_δ set in Theorem 5.3. The set $R_\Gamma \cap J$ has cardinality continuum. For each t in this set, the theorem selects a faithful projective representation with $4\chi = t$ and boundary entropy t. The stationary measure is exact dimensional, and the Ledrappier–Young identity $h_\mu = 2\chi \dim \eta$ gives $\dim \eta = 2$ [15, Theorem 2.14 and Appendix A]. Different values of t give nonisomorphic measured Γ-spaces, since Furstenberg entropy is preserved by equivariant measure-preserving isomorphisms. Each boundary is proper because its entropy is strictly smaller than h. □

The same conclusion holds for the simple random walk on F_2, with h replaced by $\frac{1}{2}\log 3$. The assertion concerns different entropy values; it makes no claim that distinct representations on one Lyapunov level yield nonisomorphic measured boundaries.

10.4. **Stable joins and endpoint convergence.** The entropy metric gives quantitative consequences for canonical factor operations. In this subsection write h for entropy, let P be the full Poisson boundary, and let $\mathbf{0}$ be the trivial factor.

Corollary 10.5. *The following hold for boundary factors of the fixed walk.*

(i) *If $\sum_n \mathsf{h}(C_n) < \infty$, $J_N = \bigvee_{n \le N} C_n$, and $J = \bigvee_{n \ge 1} C_n$, then*

$$d(J_N, J) \le \sum_{n > N} \mathsf{h}(C_n).$$

(ii) *Joins satisfy*

$$d(A \vee C, A' \vee C') \le d(A, A') + d(C, C').$$

(iii) *$d(A, P) = h - \mathsf{h}(A)$ and $d(A, \mathbf{0}) = \mathsf{h}(A)$. Thus entropy convergence to either endpoint implies convergence to that endpoint in the entropy metric and strongly for conditional expectations.*

Proof. For (i), subadditivity bounds the entropy increase in every finite tail by its total entropy; increasing-join continuity gives the same bound for J. Since $J_N \le J$, their distance is the entropy difference. For (ii), use the triangle inequality and contraction under joining a fixed factor. The identities in (iii) follow from $A \vee P = P$ and $A \vee \mathbf{0} = A$; the operator conclusion follows from (9.1). □

Corollary 8.3 supplies a further application: a scalar entropy approximation becomes exact whenever its successive join costs are summable. The density theorem provides scalar approximations, but does not by itself provide this additional control.

11. An explicit failure of representation-entropy continuity

The same rational family gives an explicit obstruction to extending entropy continuously across the real locus.

Theorem 11.1 (Exact doubling in a convergent sequence)**.** *Let*

$$z_* = \frac{1}{2} - \frac{7}{20}\sqrt{2}, \qquad s_* = \frac{1}{2} + \frac{7}{20}\sqrt{2} = 1 - z_*.$$

For the fixed uniform measure (1.1), ρ_{z_*} is a faithful algebraic representation into $\mathrm{SL}_2(\mathbb{R})$. Write $\chi_* := \chi(\rho_{z_*})$. Then $\chi_* > 0$, and there is a sequence of faithful algebraic representations ρ_{ζ_n} converging to ρ_{z_*}, with real-Zariski-dense images in $\mathrm{SL}_2(\mathbb{C})$, such that

$$h_\mu(Y_{z_*}) = 2\chi_*, \qquad \lim_{n\to\infty} h_\mu(Y_{\zeta_n}) = 4\chi_*.$$

In particular, the boundary entropy is not continuous on the locus of faithful non-elementary complex representations.

Proof. First, $7/5 < \sqrt{2} < 10/7$ gives $0 < z_* < 1/100$ and $99/100 < s_* < 1$. For a real parameter $s \in (99/100, 1)$, direct matrix identities give

$$x(s) := \operatorname{tr} A(s) = \operatorname{tr} B(s) = 2 + \frac{6s^2}{1+2s} > 2,$$

$$Z(s) := \operatorname{tr}(A(s)B(s)) = 2 + \frac{4s^2(s+2)^2}{(1+2s)^2} > 2,$$

and

$$\kappa(s) := \operatorname{tr} K(s) = 2 - \frac{32s^4(s+2)^2}{(1+2s)^2} < -28.$$

For the last bound, $(s+2)/(1+2s) > 1$ and $32(99/100)^4 > 30$. The Fricke criterion [9, Theorem 4.1.2] for a marked hyperbolic one-holed torus applies to the positive trace triple $(x(s), x(s), Z(s))$, since its commutator trace is below -2. Consequently $\langle A(s), B(s)\rangle$ is the holonomy of a marked one-holed torus with geodesic boundary corresponding to $K(s)$.

Writing $a = 1 + 14s$ and using $\det K(s) = 1$ gives

$$\det C(s) = a^2 + as\kappa(s) + s^2 < 226 - 28\left(\frac{743}{50}\right)\left(\frac{99}{100}\right) < 0.$$

Thus the real projective transformation $C(s)$ is an orientation-reversing isometry of $\mathbb{H}^2$ that centralizes the hyperbolic element $K(s)$. It fixes the two endpoints of the boundary axis individually. Such an isometry is the reflection across that axis composed with a translation along it. The displayed representation therefore doubles the one-holed torus across its geodesic boundary, with an allowed twist. The resulting closed hyperbolic surface has genus two, so ρ_s is projectively discrete and faithful. This is equivalently the standard Fuchsian combination theorem for gluing two geodesic-boundary surfaces.

In particular ρ_{s_*} is faithful. All entries of the family are rational functions with rational coefficients. The automorphism $\sqrt{2} \mapsto -\sqrt{2}$ sends s_* to z_*. It preserves all matrix relations, and therefore preserves faithfulness, including projective faithfulness. It also shows that $C(z_*)$ is invertible. Thus ρ_{z_*} is faithful and algebraic. Its image is non-elementary: faithfulness excludes a finite projective orbit, and $\operatorname{tr} A(z_*) > 2$ gives an unbounded element. Furstenberg's positivity theorem gives $\chi_* > 0$.

We next verify a uniform upper bound. For $0 < z < 1/100$,

$$\|A(z) - I\|_{\mathrm{op}} \le \|A(z) - I\|_{\mathrm{F}} \le \sqrt{13}\, z < \frac{1}{25}.$$

Here the lower-right entry of $A - I$ is $-2z(1-z)/(1+2z)$, whose absolute value is at most $2z$. The same estimate applies to B. Since these matrices have determinant one, their inverse operator norms equal their operator norms. Hence

$$\|A^{\pm 1}\|, \|B^{\pm 1}\| < \frac{26}{25}, \qquad \|K\| < \left(\frac{26}{25}\right)^4 < \frac{6}{5}.$$

Write $C = (1+14z)(I+uK)$ with $u = z/(1+14z) < 1/100$. A Neumann-series bound gives

$$\|C\|\,\|C^{-1}\| \le \frac{1+u\|K\|}{1-u\|K\|} < \frac{253}{247}.$$

Each of the eight generator matrices consequently has norm below

$$\frac{26}{25}\frac{253}{247} < \frac{11}{10}.$$

It follows by submultiplicativity that $\chi_* \le \log(11/10)$.

The quotient $a_1 \mapsto x, b_1 \mapsto y, a_2 \mapsto y, b_2 \mapsto x$ maps the fixed walk to the simple symmetric walk on F_2. Therefore

$$h := h_{\mathrm{RW}}(\Gamma, \mu) \ge \frac{1}{2}\log 3 > 4\log(11/10) \ge 4\chi_*.$$

The strict numerical comparison follows, without approximation, from $11^8 < 3 \cdot 10^8$.

It remains to produce the approximating sequence. Let V be a complex neighborhood of s_* in D on which the family remains projectively faithful, using openness of convex cocompact representations. For any small complex open set U near z_*, choose $u \in U$ and $v \in V$. Approximate $(u+v)/2$ and $(u-v)/(2\sqrt{2})$ by $a, b \in \mathbb{Q}(i)$, respectively. Then $\zeta = a + b\sqrt{2} \in U$ and its conjugate $\sigma\zeta = a - b\sqrt{2} \in V$. Thus ρ_ζ is faithful and algebraic. We can additionally arrange $\mathrm{Im}(\mathrm{tr}\, A(\zeta))^2 \neq 0$, because this is an open dense condition on the complex parameter domain. Faithfulness excludes finite projective orbits; the nonreal squared trace excludes compact image and invariant generalized circles. The representations so obtained satisfy all dynamical hypotheses of Rapaport–Ren, Theorem 1.2.

Choose $\zeta_n \to z_*$. Continuity of the Lyapunov exponent at the non-elementary limit follows from weak convergence of the unique stationary measures and Furstenberg's continuous norm-cocycle formula. The algebraic Diophantine property and the dimension formulas give

$$h_\mu(Y_{z_*}) = \min\{h, 2\chi_*\} = 2\chi_*, \qquad h_\mu(Y_{\zeta_n}) = \min\{h, 4\chi(\rho_{\zeta_n})\}.$$

The real formula uses [12, Theorems 1.1 and 3.4]. The complex formula uses [15, Theorems 1.2 and 2.14], together with the identity of Furstenberg entropy and the first-step mutual information $H(p) - H(\omega_0 \mid L)$. Since $4\chi_* < h$, the second expression converges to $4\chi_*$, as claimed. □

The limiting representation preserves a generalized circle. Thus this example disproves entropy continuity on the faithful non-elementary locus; it does not assert a discontinuity with both the sequence and its limit inside the real-Zariski-dense locus. All the stationary spaces here are genuine boundary factors. Weak convergence of their stationary measures follows from uniqueness at the non-elementary limit, so the example directly separates weak convergence from entropy convergence.

12. Levelwise saturation and holomorphic rigidity

12.1. **A limitation of semicontinuity and bifurcation support.** Full realization by the projective family would follow from finding, for each $0 < t < h$, a parameter with $4\chi = t$ and a stationary measure of dimension 2. The generic realization theorem establishes this for a

comeager set of targets. The following example shows that lower semicontinuity, dense saturation, and full bifurcation support alone do not imply saturation on every level.

Let $\Omega = \{r < |z| < R\}$, where $0 < r < R < \sqrt{h}$, and put $p(z) = |z|^2$. Given any countable dense set $A \subset \Omega$, choose $c \in p(\Omega) \setminus p(A)$ and $0 < \varepsilon < c$. Define

$$F(z) = p(z) - \varepsilon \mathbf{1}_{\{p(z)=c\}}.$$

Then $0 \le F \le p$, F is lower semicontinuous, and $p - F$ is upper semicontinuous. Equality $F = p$ holds on A and on an open dense set of full planar measure. Moreover, p is smooth and strictly subharmonic, and $dd^c p$ has full support. Nevertheless, equality fails everywhere on the level circle $\{p = c\}$, which has positive logarithmic capacity. Thus these properties neither ensure saturation on each level nor force the exceptional set to have zero capacity. The example concerns the logical strength of the stated hypotheses; it does not assert that this defect occurs for Furstenberg entropy.

12.2. **Holomorphic families with constant Lyapunov exponent.**

Proposition 12.1 (Rigidity on a subcritical constant-Lyapunov disk)**.** *Let $(\rho_w)_{w\in\Delta}$ be a holomorphic family of representations of Γ into $\mathrm{PSL}_2(\mathbb{C})$, with ρ_0 faithful and non-elementary. If*

$$\chi(\rho_w) = c < h/4 \qquad (w \in \Delta),$$

then, after shrinking Δ, all representations in the family are conjugate.

Proof. Shrink the disk so that every representation is non-elementary. Faithfulness at zero implies general faithfulness: for each nontrivial word, its identity locus is a proper analytic subset. If the family were nontrivial up to conjugacy, it would be admissible for [5, Theorem 3.9]; finite support of μ supplies the exponential moment condition. Constancy of χ makes $dd^c\chi$ vanish. The support theorem and [5, Theorem 2.6] would then make the family discrete and faithful. The packing inequality $h \le 4\chi = 4c$ contradicts the hypothesis. □

The proposition applies to holomorphic curves in multiparameter representation spaces as well. Thus a holomorphic deformation through a faithful subcritical representation that stays on one Lyapunov level can vary only by conjugacy, which preserves boundary entropy. The existence and regularity of real paths within such levels remain separate questions: continuity or Hölder continuity of χ alone does not supply an analytic implicit-function theorem.

13. Open realization questions

For the fixed genus-two walk, the established exact interval is $[0, \frac{1}{2}\log 3]$, and the complement of the boundary entropy spectrum in $(0, h)$ is meager. Two further selection statements would resolve individual missing values.

Question 13.1 (Saturation on each Lyapunov level)**.** For every $0 < t < h$, does there exist a projectively faithful parameter $z \in Q$ with

$$4\chi(z) = t, \qquad \dim \eta_z = 2?$$

An affirmative answer would give full realization by (3.1). Proposition 4.1 supplies faithful points on every required level, while Lemma 5.2 supplies dimension-two saturation on a residual set of parameters. The missing assertion is their intersection on every level. The example in Section 12 shows that lower semicontinuity, dense saturation, and full support of the bifurcation current do not alone prove it. Dense kernel specializations and unbounded word contact also prevent a uniform finite-order transversality argument on the subcritical region.

The boundary lattice gives a different formulation. For a factor A, set $r_A(C) = \mathsf{h}(C) - \mathsf{h}(A \wedge C)$, where h denotes boundary entropy.

Question 13.2 (Arbitrarily small relative extensions)**.** Does every factor A with $\mathsf{h}(A) < h/2$ satisfy

$$\inf_{C \not\leq A} r_A(C) = 0?$$

Does the same assertion hold for every proper factor A?

Proposition 9.3 proves that the first assertion would give $[0, h/2] \subseteq \mathscr{H}_{\partial}(\Gamma, \mu)$; the second would give full realization. The common-factor term allows C to have substantial absolute entropy if most of it is already present in A. Neither the density theorem nor completeness of the entropy metric currently provides such an extension. The metric criterion instead asks for a choice of approximants whose joins introduce a summable amount of additional entropy. These are precise remaining requirements, rather than consequences of scalar density.

Appendix A. An elementary continuity argument for symmetric walks

For a finitely supported symmetric measure in $\mathrm{SL}_2(\mathbb{C})$, $\chi \geq 0$ and

$$\chi = \inf_{n \geq 1} \frac{1}{n} \sum_{s_1, \ldots, s_n} \mu(s_1) \cdots \mu(s_n) \log \|\rho(s_1 \cdots s_n)\|.$$

The right-hand side is an infimum of continuous functions of the matrix tuple, proving upper semicontinuity. At a non-elementary noncompact image, uniqueness of the stationary projective measure implies its weak continuity: any subsequential limit remains stationary for the limiting tuple. The Furstenberg integral formula, whose integrand is jointly continuous on the compact projective space, then gives continuity of χ at that tuple. At a relatively compact image, $\chi = 0$. At an image fixing one projective point, conjugate to upper triangular matrices. The two diagonal exponents are the averages of opposite log-modulus characters; inversion symmetry makes both zero. The Lyapunov exponents of a triangular cocycle are its diagonal exponents, so $\chi = 0$. If precisely a two-point set is preserved, conjugate to monomial matrices. If the two points are not interchanged, the same argument applies. Otherwise the sign process on the two coordinate axes is a finite irreducible Markov chain with uniform stationary law; the two opposite logarithmic increments have zero averaged drift, again giving $\chi = 0$. A finite orbit of size at least three forces a finite projective image and hence $\chi = 0$. Upper semicontinuity and nonnegativity now give continuity at all remaining tuples. This proof uses no entropy regularity assertion.

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